

Dedalus_ECS Documentation

Van Duc Nguyen^{1,*} and Chang Liu^{1,†}

¹*School of Mechanical, Aerospace, and Manufacturing Engineering,
University of Connecticut, Storrs, CT 06269, USA*

(Dated: February 10, 2026)

CONTENTS

I. How to run dedalus_ecs	2
A. Finding equilibrium states	2
B. Finding traveling waves	2
C. Finding periodic orbits	2
D. Finding relative periodic orbits	2
II. Governing equations	2
III. Computation of ECS and their verification	2
A. ECS classification	2
1. Equilibrium states	3
2. Periodic orbits	3
3. Traveling waves and Relative periodic orbits	3
B. Newton-Raphon method	4
C. Hookstep modification	4
D. Linear system	4
IV. Symmetry	4
A. Reflection symmetries	4
B. Translation symmetries	4
V. Linear stability	4
A. Linear operator	4
B. Arnoldi iteration	4
C. Implementation	4
D. Zero eigenmodes due to neutral translation	4
VI. Verifications and validations	4
VII. Parametric continuation	4
A. Arc-length continuation method	4
B. Phase condition	4
References	4

* duc.nguyen@uconn.edu

† chang_liu@uconn.edu

I. HOW TO RUN DEDALUS_ECS

A. Finding equilibrium states

B. Finding traveling waves

C. Finding periodic orbits

D. Finding relative periodic orbits

II. GOVERNING EQUATIONS

We model the vertically periodic salt finger convection in terms of the velocity $\mathbf{u}(\mathbf{x}, t) = (u, w)$, temperature scalar $T(\mathbf{x}, t)$ and salinity scalar $S(\mathbf{x}, t)$, where $\mathbf{x} = (x, z)$ and t are the partial coordinates and the time. The domain is a fully periodic 2D rectangle box of $\mathbf{x} \in [0, L_x] \times [0, L_z]$. We employ the non-dimensional governing equations

$$\nabla \cdot \mathbf{u} = 0, \quad (1a)$$

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + Pr \nabla^2 \mathbf{u} + Pr Ra (T - R_\rho^{-1} S) \hat{\mathbf{e}}_z, \quad (1b)$$

$$\partial_t T + \mathbf{u} \cdot \nabla T = \nabla^2 T, \quad (1c)$$

$$\partial_t S + \mathbf{u} \cdot \nabla S = \tau \nabla^2 S, \quad (1d)$$

where p is the pressure and $\hat{\mathbf{e}}_z$ is the vertical unit vector. The equations were constructed through the Oberbeck–Boussinesq (OB) approximation with non-dimensionalization of thermal diffusion scale $[\cdot]$. The resulting system is governed by the dimensionless parameters: Rayleigh number Ra , Prandtl number Pr , density ratio R_ρ and the diffusivity ratio τ . We fix parameter set at $Ra = 10^5$, $Pr = 7$, $R_\rho = 40$, $\tau = 0.01$ and domain size $L_x = 0.4, 0.6, 0.8$ and $L_z = 1$ with grid mesh $N_x \times N_z = 128 \times 128$. The dealias factor is $3/2$. The equations 2 are solved by open-source pseudospectral code Dedalus [1].

To determine ECS of the system, we use the shooting-based Newton-Raphson method [2] coupled with matrix-free Krylov method to find the state vector $\xi = [u, w, \theta, s]^T$ including phase space dimensions $(3/2)^2 \times 128 \times 128$. Here, fluctuations of temperature θ and salinity s around a trivial state $T = S = z$ are used to simplify the equations 2. State vectors are considered as ECS when $\xi - \sigma \mathcal{F}^T(\xi) = \mathbf{0}$, where σ denotes the symmetry operator and $\mathcal{F}^T$ is the time- T forward integration. We employ the hook-step trust-region optimization [3] and matrix-free generalized minimum residual algorithm [4] to improve the convergence.

$$\nabla \cdot \mathbf{u} = 0, \quad (2a)$$

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + Pr \nabla^2 \mathbf{u} + Pr Ra (T - R_\rho^{-1} S) \hat{\mathbf{e}}_z, \quad (2b)$$

$$\partial_t T + \mathbf{u} \cdot \nabla T = \nabla^2 T, \quad (2c)$$

$$\partial_t S + \mathbf{u} \cdot \nabla S = \tau \nabla^2 S, \quad (2d)$$

III. COMPUTATION OF ECS AND THEIR VERIFICATION

A. ECS classification

The velocity $\mathbf{u} = (u, w)$, temperature fluctuation θ , and salinity fluctuation s are collected into a single state vector $\xi = [u, w, \theta, s]^T$ including a phase space dimension of $(3/2)^2 \times N_x \times N_z$. ECS satisfies the following condition

$$\xi - \sigma \mathcal{F}^T(\xi) = \mathbf{0}, \quad (3)$$

where σ denotes the symmetry operator and $\mathcal{F}^T$ is the time- T forward integration by Dedalus. We use a residual of 10^{-10} to determine the numerical convergence of ECS. The ECS determined by equations 3 can be classified into various solution types:

- (1) Equilibrium state (EQ): the state $\xi(x, z, t)$ remains constant in time as $T \rightarrow \infty$.
- (2) Traveling wave (TW): the state $\xi(x, z, t)$ undergoes spatial translation within the domain with a specific phase speed (c_x, c_z) .

(3) Periodic orbit (PO): the state $\xi(x, z, t)$ returns after a finite time T .

(4) Relative periodic orbit (RPO): the state $\xi(x, z, t)$ returns after a finite time T at a shifted domain $(x + d_x$ and/or $z + d_z)$.

To solve the equation 3, we employ the Newton-Raphson iteration method.

1. Equilibrium states

Equilibrium is a common ECS which requires

$$\partial_t \xi = \mathbf{0}. \quad (4)$$

Based on this property, a equilibrium also satisfy the equation 3 with a free T . For convenience and efficiency, we fix $T = 0.2$ in all computation of EQ and later TW. This values requires a typical number $\mathcal{O}(10^3)$ of time step in the present system [? ?]. The symmetry operator is used as a option in our code to find EQ of specific symmetry.

2. Periodic orbits

The equation 3 handles exactly the definition of a periodic orbit that will return after exactly a time interval T . In this case, T represents the time period of PO. Unlike the equilibrium state, T is unknown finite value that needs to be determined for a PO. Therefore, finding PO requires adding one more additional variable for time period T . A temporal phase condition

$$\langle \xi - \xi_{\text{ref}}, \partial_t \xi_{\text{ref}} \rangle = 0, \quad (5)$$

is used to optimize T .

3. Traveling waves and Relative periodic orbits

spatial phase condition

inner product

$$\langle \xi - \xi_{\text{ref}}, \partial_x \xi_{\text{ref}} \rangle = 0, \quad (6a)$$

$$\langle \xi - \xi_{\text{ref}}, \partial_z \xi_{\text{ref}} \rangle = 0, \quad (6b)$$

B. Newton-Raphon method

C. Hookstep modification

D. Linear system

IV. SYMMETRY

A. Reflection symmetries

B. Translation symmetries

V. LINEAR STABILITY

A. Linear operator

B. Arnoldi iteration

C. Implementation

D. Zero eigenmodes due to neutral translation

In salt finger convection, there are neutral drifts by continuous translation symmetry ($c_x \partial_x \xi$ and/or $c_z \partial_z \xi$ with tiny neutral phase speeds c_x, c_z). This often leads to near-zero unstable eigenvalues in linear stability problem. To remove these zero eigenvalues, we project out the spatial derivatives ($\partial_x \xi$ and $\partial_z \xi$) of linearization operator.

VI. VERIFICATIONS AND VALIDATIONS

VII. PARAMETRIC CONTINUATION

To conduct the parametric continuation, we employ the arc-length continuation strategy which have been applied successfully in Channelflow 2.0 [1]. This strategy enable to predict the fold continuation of ECS in the parameter space.

A. Arc-length continuation method

B. Phase condition

-
- [1] K. J. Burns, G. M. Vasil, J. S. Oishi, D. Lecoanet, and B. P. Brown, Phys. Rev. Research **2**, 023068 (2020).
 - [2] C. T. Kelley, *Solving nonlinear equations with Newton's method* (SIAM, 2003).
 - [3] D. Viswanath, J. Fluid Mech. **580**, 339 (2007).
 - [4] G. J. Chandler and R. R. Kerswell, J. Fluid Mech. **722**, 554 (2013).